

An Edge Companion and Fall-Detection Desktop Robot for Older Adults Living Alone

Your Name, Teammate Name

Computer Organization and Architecture
Your University

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1 Topic

This project develops a compact desktop companion robot for older adults living alone, including left-behind older adults in rural communities. The robot runs on a Raspberry Pi 5 and combines voice interaction, a small local language model, an OLED status display, visual person tracking, a two-axis gear-driven head, and vision-based fall detection. Its purpose is to provide a more available daily interaction channel and to issue a local alert when a probable fall is detected within the camera view. It is not a medical device, does not diagnose any condition, and makes no clinical claim about loneliness, mental health, or fall prevention.

2 Introduction

Older adults who live alone can face long periods without routine social interaction and may have limited access to immediate assistance after an accident. A conventional desktop voice assistant can offer reminders and conversation, but it does not orient its display, camera, or lamp toward the person using it. A small desktop robot can make the interaction more present by turning toward the current user while also serving as an always-on safety-oriented sensing platform for its visible area.

The project is motivated by a social need rather than a clinical intervention claim. We aim to build a practical prototype that can greet and converse with a user, turn its camera and lamp head toward the user on a desk, and alert locally when a probable fall occurs within the camera view. The computer-architecture challenge is to execute these concurrent workloads—speech processing, lightweight language generation, vision inference, motor control, and display—within the memory, power, and thermal limits of a Raspberry Pi 5.

3 Related Approaches

Socially assistive robots have been studied as a way to support communication and daily activities for older adults, while their role must remain carefully bounded and user-centered [1]. Existing fall-detection systems commonly use wearables, ambient sensors, or cameras. Vision avoids requiring the user to remember a wearable, but robust detection must distinguish a true fall from sitting, bending, or lying down [2]. We therefore use a temporal pose-based decision rather than classifying a single frame.

For perception, a detector and tracker maintain one temporary interaction target; ByteTrack is a representative method for associating detections across frames [3]. For dialogue, a 4-bit quantized Qwen2.5-0.5B model is suitable for short, constrained edge-side responses [4]. These approaches are usually studied separately. Our proposed contribution is an integrated edge system that schedules them together and evaluates both useful behavior and resource cost.

4 Proposed Solution

The robot uses a Raspberry Pi 5 with 4 GB RAM, active cooling, an RGB-D camera, USB microphone and speaker, an I2C OLED, two operational-amplifier control stages, two worm-gearred motors, and two gear trains. It is a compact desktop device and has no floor chassis or wheels. The Pi executes perception, dialogue, high-level control, and display services. Its pan and tilt commands are passed to the two control stages, which generate the control signals for the two worm-gearred motor channels and mechanically turn the camera/lamp head.

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RGB-D camera -> person detector/tracker -> pose sequence
                |-> temporal fall decision -> local alert
                |-> target image position -> pan/tilt command
                |-> two control stages -> worm-gearred motors
                |-> two-axis camera/lamp head
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Microphone -> wake word -> ASR -> local Qwen 0.5B -> TTS -> speaker
OLED <- time/date/weather/system state/fall-alert state
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The vision pipeline does not perform face recognition. It tracks only the current interaction target and uses depth, torso orientation, and the change of body-keypoint height across a short frame window to detect a fall candidate. A fall alert is issued only after the temporal decision is stable; this reduces alarms caused by brief bending or sitting. For user-oriented interaction, horizontal and vertical image-center errors are translated into pan and tilt angle commands. The worm-gearred mechanism gives the desktop head stable, low-speed movement; target loss produces a hold-position command. Because the device stays on a desk, fall detection is limited to the camera’s visible area.

The companion pipeline is intentionally non-clinical. A wake word activates ASR, a short local Qwen response, and TTS. Dialogue is limited to greetings, simple daily prompts, reminders, and approved non-clinical information. The system does not offer diagnosis, therapy, or emergency triage. Raw video and audio are not retained by default; the public repository will contain only code, hardware schematics, synthetic examples, and de-identified configuration data.

Table 1: Prototype implementation choices.

Component	Proposed choice
Compute platform	Raspberry Pi 5, 4 GB RAM, active cooling
Dialogue	Wake word, ASR, 4-bit Qwen2.5-0.5B, TTS
Perception	RGB-D camera, person tracking, pose sequence, fall decision
Motion mechanism	Two worm-gearred motors and gear trains for pan/tilt
Control stages	Two operational-amplifier control stages for the motor channels
Safety	Hold position on target loss; local fall alert only in camera view
User display	I2C OLED for time, status, and alerts

5 Evaluation

We will evaluate the prototype with repeatable indoor experiments rather than claim clinical benefit. First, the fall module will be tested on public fall-action video data and controlled, consented demonstrations of walking, sitting, bending, lying down, and simulated falls. We will report fall recall, precision, false alarms per hour, and detection-to-alert latency. The proposed temporal pose-and-depth decision will be compared with a single-frame bounding-box-height heuristic; the expected advantage is fewer false alarms during ordinary actions.

Second, we will measure active desktop tracking with one moving target in front of the robot. Metrics are target-in-frame ratio, image-center error, pan/tilt response time, and target reacquisition time. A fixed-head camera provides a baseline for showing whether the gear-driven two-axis head keeps the interaction target visible more reliably.

Third, we will benchmark edge execution on the Raspberry Pi 5: vision frame rate, end-to-end fall-alert latency, ASR-to-first-audio latency, CPU load, RAM use, board temperature, and power draw. Finally, a privacy audit will verify that raw video/audio, identity mappings, and sensitive data are absent from the repository and are not stored by default on the device.

6 Conclusion

This proposal presents an edge companion and fall-detection desktop robot for older adults living alone. Its value is social and practical: it makes routine interaction more available and provides a local safety alert when a probable fall occurs in the camera view, without claiming medical or clinical efficacy. By combining pose-based temporal fall detection, gear-driven pan/tilt interaction, local voice interaction, and resource benchmarks on a Raspberry Pi 5, the project directly connects a socially meaningful application to computer-organization and architecture concerns such as scheduling, latency, memory, power, and embedded control.

References

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